GOVERNMENT PURCHASES MULTIPLIER: THEORY II

Fiscal and Monetary Policy 2026

Piotr Żoch

February 21, 2026

BEYOND THE NGM

TOWARDS HIGHER MULTIPLIERS?

- We saw that in the NGM short-run multipliers tended to be small.
- We will now discuss to what extent this result changes when we move to a setup with nominal rigidities (New Keynesian models).
- We then move to an even more general setup that allows for heterogeneous agents and incomplete markets on top of nominal rigidities.

ROADMAP (WOODFORD 2011)

- Woodford (2011) analyzes the multiplier in a series of simple models to isolate the key mechanisms.
- **Step 1**: Neoclassical benchmark (no capital, flexible prices) → multiplier < 1.
- **Step 2**: Sticky prices + constant real interest rate → multiplier = 1.
- **Step 3**: Sticky prices + Taylor rule → multiplier < 1 (but > neoclassical).
- **Step 4**: Sticky prices + zero lower bound → multiplier > 1.
- Key insight: the monetary policy response is the crucial determinant of the multiplier in NK models.

STEP 1: NEOCLASSICAL BENCHMARK

- Economy with no capital – essentially static. Production: $y_t = f(n_t)$. Resource constraint: $y_t = c_t + g_t$.
- Household and firm optimization requires

$$\frac{v'(n_t)}{u'(c_t)} = \frac{W_t}{P_t}, \quad \frac{W_t}{P_t} = f'(n_t).$$

- Combining and substituting $c_t = f(n_t) - g_t$, we get a single equation in y_t:

$$u'(y_t - g_t) = \tilde{v}'(y_t), \quad \text{where } \tilde{v}(y) \equiv v\left(f^{-1}(y)\right).$$

STEP 1: NEOCLASSICAL BENCHMARK

- Implicit differentiation of $u'(y-g) = \tilde{v}'(y)$ yields the multiplier:

$$\frac{dy}{dg} = \frac{\eta_u}{\eta_u + \eta_v} \equiv \Gamma < 1,$$

  where $\eta_u \equiv -\bar{y}\, u''/u' > 0$ and $\eta_v \equiv \bar{y}\, \tilde{v}''/\tilde{v}' > 0$.

- Government purchases always crowd out private expenditure:

$$dc = dy - dg = -(1-\Gamma)\, dg < 0.$$

- This holds regardless of how g is financed (lump-sum taxes, Ricardian equivalence).
- Key: what can change the multiplier is something that breaks the intratemporal condition:

$$u'(y_t - g_t) = \tilde{v}'(y_t).$$

STEP 2: CONSTANT REAL INTEREST RATE

- Now suppose prices (or wages) are sticky and monetary policy maintains the real rate constant at $r = \beta^{-1} - 1$.
- The Euler equation:

$$\frac{u'(c_t)}{u'(c_{t+1})} = \beta\left(1 + r_{t+1}\right).$$

- With $r_{t+1} = \bar{r}$ for all t:

$$\frac{u'(c_t)}{u'(c_{t+1})} = 1 \implies c_t = c_{t+1} \text{ for all } t.$$

- For temporary changes in g_t (with $g_t \to \bar{g}$ in the long run), consumption is constant at the steady-state level $\bar{c} = \bar{y} - \bar{g}$.

STEP 2: CONSTANT REAL INTEREST RATE

- Since $y_t = c_t + g_t$ and $c_t = \bar{c}$:

$$y_t = \bar{c} + g_t \implies \frac{dy_t}{dg_t} = 1.$$

- No crowding out of private expenditure! But no stimulus of additional private expenditure either.
- This result is independent of the details of price/wage adjustment – it depends only on the Euler equation and the resource constraint.
- It holds regardless of how the government finances its spending (not a balanced-budget result).
- This is what happens in a large class of representative agent New Keynesian models (RANK).

STEP 2: WHY DO NOMINAL RIGIDITIES MATTER?

- With flexible prices and wages, the intratemporal condition pins down c_t given g_t:

$$\tilde{v}'(c_t + g_t) = u'(c_t).$$

  The Euler equation then just determines the real rate.

- This condition combines two optimality conditions:
  1. Household: $v'(n_t) = \frac{W_t}{P_t} u'(c_t)$ (labor-leisure choice).
  2. Firm: $f'(n_t) = \frac{W_t}{P_t}$ (labor demand).
- With nominal rigidities, at least one of these fails as an equality → the intratemporal condition no longer pins down c_t.
- Instead, the Euler equation (+ monetary policy) determines consumption, and an intratemporal wedge adjusts.

STEP 2: WHY DO NOMINAL RIGIDITIES MATTER?

- Example: nominal wages are rigid at $\bar{W}$ and the household supplies any labor demanded at that wage.
- The household's labor-leisure FOC no longer holds. Instead, we can write:

$$\frac{v'(n_t)}{u'(c_t)} = \frac{\bar{W}}{P_t} \cdot \mu_t^W,$$

  where $\mu_t^W \neq 1$ is a wage markup (wedge) that adjusts endogenously.
- The intratemporal condition no longer constrains c_t – it just determines the wedge μ_t^W residually.
- With sticky prices the logic is analogous, but the wedge is on the firm's side:

$$\bar{P} \cdot f'(n_t) = W_t \cdot \mu_t^P,$$

  where μ_t^P is the price markup.

STEP 2: WHY DO NOMINAL RIGIDITIES MATTER?

- Nominal rigidities mean that someone commits to a price and then meets demand at that price.
- Output is demand-determined: $y_t = c_t + g_t$, where c_t is pinned down by the Euler equation and monetary policy.
- By controlling real interest rates, the central bank chooses the level of demand in the economy.

STEP 3: TAYLOR RULE

- More realistic: the central bank follows a Taylor rule,

$$i_t = \bar{r} + \phi_\pi \pi_t + \phi_y \hat{y}_t, \quad \phi_\pi > 1, \ \phi_y > 0.$$

- Government purchases increase output → increase inflation → central bank raises the real interest rate.
- Higher real rate → consumption falls → multiplier is less than 1.
- But still larger than the neoclassical benchmark Γ:

$$\Gamma < \gamma_y < 1.$$

- The more aggressive the monetary response (higher ϕ_π, ϕ_y) or the more flexible prices, the smaller the multiplier.

STEP 4: ZERO LOWER BOUND

- When the nominal interest rate is at the zero lower bound: $i_t = 0$.
- Suppose that even with the fiscal expansion, the central bank keeps the nominal rate at zero.
- Fiscal expansion raises expected inflation → the real interest rate falls:

$$r_t = i_t - E_t\pi_{t+1} = -E_t\pi_{t+1} < \bar{r}.$$

- Lower real rate → consumption increases → multiplier is greater than 1.
- This is the case where fiscal stimulus is most effective.
- Important caveat: the increase in g should not persist beyond the period when the ZLB binds.

WOODFORD (2011): MULTIPLIER AT THE ZLB

- Figure shows the multiplier dY/dG as a function of the persistence of government spending at the ZLB (μ).
- As $\mu \to \bar{\mu}$, the multiplier grows without bound.
- The multiplier is large precisely when the output cost of the ZLB episode is large.

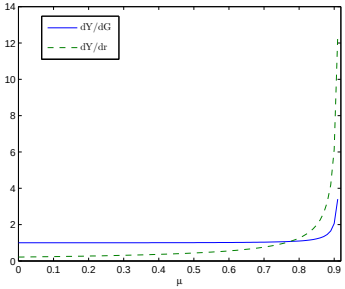


Figure 2: Derivatives of Y_L with respect to the values of r_L and G_L, for alternative assumed degrees of persistence μ of the financial disturbance. Other parameter values are taken from Eggertsson (2009).

Source: Woodford (2011), Figure 2.

WOODFORD (2011): TIMING OF FISCAL STIMULUS MATTERS

- Now vary λ: the fraction of spending that persists after the ZLB episode ends.
- If spending persists beyond the ZLB, the multiplier drops below 1 and can even become negative.
- Key lesson: fiscal stimulus should be temporary and concentrated during the ZLB period.

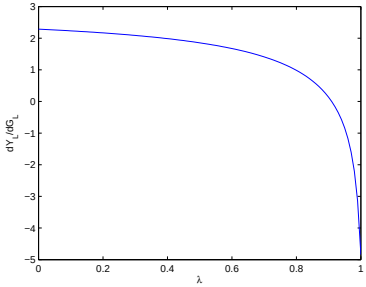


Figure 3: Derivative of Y_L with respect to G_L, for alternative degrees of persistence λ of the fiscal stimulus after the end of the financial disturbance. Other parameter values are taken from Eggertsson (2009).

Source: Woodford (2011), Figure 3.

SUMMARY: WOODFORD (2011)

- The multiplier depends crucially on the monetary policy response:

Monetary policy	**Multiplier**
Flexible prices (neoclassical)	$\Gamma < 1$
Strict inflation targeting	$\Gamma < 1$
Taylor rule	$\Gamma < \gamma_y < 1$
Constant real rate	$= 1$
Zero lower bound	> 1

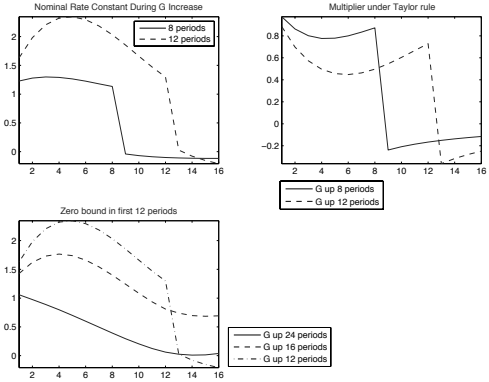
- Nominal rigidities are necessary but not sufficient for a large multiplier – monetary accommodation is key.
- Note: it was a New *Keynesian* model, but we have not talked about marginal propensities to consume at all! The real rate was the key!

WHAT ABOUT CAPITAL? (CHRISTIANO, EICHENBAUM, REBELO 2011)

- Woodford's analysis assumed no capital. How does capital accumulation change the results?
- **Outside the ZLB**: the multiplier with capital is ≈ 0.9.
  - Higher g raises the real rate $\rightarrow$ crowds out both consumption and investment.
- **At the ZLB**: the multiplier with capital is ≈ 3.9 (impact).
  - At the ZLB the real rate rises (deflationary spiral) $\rightarrow$ investment falls sharply.
  - Fiscal stimulus closes the saving-investment gap $\rightarrow$ the multiplier is much larger.
- The gap between ZLB and non-ZLB multipliers is enormous: 0.9 vs 3.9.
- Bottom line: capital reinforces the key message – the multiplier is large precisely when the ZLB costs are large.

CER (2011): MULTIPLIER IN A MEDIUM-SIZE DSGE MODEL

- ACEL model with sticky wages/prices, variable capital utilization, investment adjustment costs.
- **Taylor rule**: multiplier ≈ 0.7–1.0, decreasing with persistence.
- **ZLB**: multiplier ≈ 1.2–2.3, increasing with ZLB duration.
- Spending that comes after the ZLB dulls the impact.



TAKING STOCK

- In the models we have studied so far, intertemporal consumption-savings decisions of households were crucial.
- The (representative) household was always on the Euler equation.
- These models were Ricardian, so we were unable to talk about different effects of deficit- vs tax-financed spending.
- We will now move to a more general setup.

AUCLERT, ROGNLIE, AND STRAUB (2024): THE INTERTEMPORAL KEYNESIAN CROSS (IKC)

IKC

- Key insight: intertemporal marginal propensities to consume (iMPCs) are sufficient statistics for the effect of fiscal policy.
- Fiscal multipliers depend on the interaction of public deficits and iMPCs.
- If we want to know the effect of fiscal policy, we should measure iMPCs in the data - or at least use models that generate iMPCs roughly consistent with the data.

HOUSEHOLD

- Household i solves:

$$\max_{\{c_{i,t},b_{i,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(u\left(c_{i,t}\right) - \nu\left(h_{i,t}\right)\right)$$

$$b_{i,t} + c_{i,t} = \left(1 + r_t^B\right) b_{i,t-1} + z_{i,t} \left(1 - \tau_t\right) \frac{W_t}{P_t} h_{i,t}$$

$$b_{i,t} \geq \underline{b}$$

- $b_{i,t}$ are real assets, r_t^B is the real interest rate, P_t is the price level, W_t is the nominal wage,
- $z_{i,t}$ is the idiosyncratic shock to income, $\mathbb{E}\left[z_{i,t}\right] = 1$.
- $h_{i,t}$ taken as given - households committed to supply any quantity that is demanded (labor unions in the background).

PRODUCTION

- A representative firm solves:

$$\max_{h_t} \quad P_t y_t - W_t h_t$$

$$y_t = A_t h_t$$

- h_t is a bundle of labor supplied by labor unions $\ell \in [0,1]$, $h_t = \left(\int_0^1 h_{\ell,t}^{\frac{\sigma_W - 1}{\sigma_W}} d\ell \right)^{\frac{\sigma_W}{\sigma_W - 1}}$.
- Each union supplies $h_{\ell,t} = \int_0^1 z_{i,t} h_{i,\ell,t} d\ell$ with the allocation rule $h_{i,\ell,t} = l(z_{i,t}) h_{\ell,t}$ such that $\int z_{i,t} l(z_{i,t}) di = 1$. We have $h_{i,t} = \int_0^1 h_{i,\ell,t} d\ell$.
- Unions set nominal wage growth to maximize HH welfare subject to adjustment costs $\rightarrow$ there is some wage Phillips curve.

GOVERNMENT

- The government sets bounded paths of government purchases g_t, tax revenue T_t, and government debt b_t^G that satisfy:

$$b_t^G = \left(1 + r_t^B\right) b_{t-1}^G + g_t - T_t$$

- Tax revenue $T_t = \tau_t \cdot y_t$.
- We assume the government controls the real interest rate r_t^B directly.
- Interpretation: there is a nominal interest rate rule that sets $i_t = r_t^B + E_t \pi_{t+1}$, where π_t is inflation.
- This allows us to sidestep the discussion of inflation dynamics.

EQUILIBRIUM

- Sequences of prices and allocations such that agents optimize, government budget constraint holds, and markets clear in every period:

$$\int c_{i,t} di + g_t = y_t,$$
$$\int b_{i,t} di = b_t^G.$$

INCOME DISTRIBUTION

- Recall that individual income is $z_{i,t}\left(1-\tau_t\right)\frac{W_t}{P_t}h_{i,t}$.
- Optimality condition of the firm implies:

$$\frac{W_t}{P_t} = A_t \rightarrow \frac{W_t}{P_t}h_t = y_t$$

- With symmetric unions:

$$h_{i,t} = l\left(z_{i,t}\right)h_t$$

- Equilibrium after-tax income of household i is:

$$z_{i,t}\left(1-\tau_t\right)\frac{W_t}{P_t}h_{i,t} = z_{i,t}l\left(z_{i,t}\right)\left(y_t - T_t\right).$$

HOUSEHOLD BLOCK

- Each household takes as given:
  - The sequence of real interest rates $\left\{r_t^B\right\}_{t=0}^{\infty}$.
  - The sequence of aggregate income net of taxes $\left\{y_t - T_t\right\}_{t=0}^{\infty}$.
  - Process for idiosyncratic productivity $z_{i,t}$.
  - Its initial wealth $b_{i,-1}$.

1. Solve the household problem for each household $i \rightarrow$ policy functions
2. Policy functions + idiosyncratic productivity process $\rightarrow$ evolution of distribution
3. Evolution of distribution + policy functions $\rightarrow$ aggregate consumption and savings.

HOUSEHOLD BLOCK

- We obtain aggregate functions

$$\mathcal{B}_t\left(\left\{r_s^B, y_s, T_s\right\}_{s=0}^{\infty}\right), \quad \mathcal{C}_t\left(\left\{r_s^B, y_s, T_s\right\}_{s=0}^{\infty}\right)$$

  such that

$$\int b_{i,t} di = \mathcal{B}_t\left(\left\{r_s^B, y_s, T_s\right\}_{s=0}^{\infty}\right), \quad \int c_{i,t} di = \mathcal{C}_t\left(\left\{r_s^B, y_s, T_s\right\}_{s=0}^{\infty}\right).$$

  We suppress the dependence on the initial conditions.

- Household block: $\boldsymbol{y}, \boldsymbol{T}, \boldsymbol{r} \rightarrow \boldsymbol{c}, \boldsymbol{b}$
- These functions embed the notion of household (and firm!) optimization.

EQUILIBRIUM

- Market clearing conditions can be written as

$$y_t = \mathcal{C}_t\left(\left\{r_s^B, y_s, T_s\right\}_{s=0}\right) + g_t,$$
$$b_t^G = \mathcal{B}_t\left(\left\{r_s^B, y_s, T_s\right\}_{s=0}\right).$$

- Using the government budget constraint to solve for b_t^G and applying Walras' Law we can find general equilibrium allocations given government policies $\{g_t, T_t\}_{t=0}^{\infty}$ as $\{y_s\}_{t=0}^{\infty}$ that solves the system of equations.

$$y_t = \mathcal{C}_t\left(\left\{r_s^B, y_s, T_s\right\}_{s=0}\right) + g_t$$

EQUILIBRIUM

- In a stacked form:

$$\underbrace{\boldsymbol{y}}_{\text{aggregate income}} = \underbrace{\mathcal{C}\left(\boldsymbol{r}^{\boldsymbol{B}}, \boldsymbol{y}, \boldsymbol{T}\right) + \boldsymbol{g}}_{\text{aggregate demand}} .$$

FISCAL POLICY

- We will now consider the effect of fiscal policy on the economy around the steady state.
- Notation: $d\boldsymbol{g} = \{dg_t\}_{t=0}^{\infty}$ and $d\boldsymbol{T} = \{dT_t\}_{t=0}^{\infty}$.
- Assume monetary policy is such that the real rate is constant at r^B.
- Focus on policies with PDV = 0, i.e. such that

$$\sum_{t=0}^{\infty} \left(\frac{1}{1+r^B} \right)^t (dg_t - dT_t) = 0$$

RESPONSE TO FISCAL POLICY

- Totally differentiate the goods market clearing condition around the steady state to get:

$$dy_t = \sum_{s=0}^{\infty} \left[\frac{\partial \mathcal{C}_t}{\partial y_s} dy_s + \frac{\partial \mathcal{C}_t}{\partial T_s} dT_s \right] + dg_t$$

- Let $\mathbf{C}_y$ be the Jacobian of $\mathcal{C}$ with respect to $\boldsymbol{y}$ and $\mathbf{C}_T$ the Jacobian of $\mathcal{C}$ with respect to $\boldsymbol{T}$: $\mathbf{C}_{y_{t,s}} = \frac{\partial \mathcal{C}_t}{\partial y_s}$, $\mathbf{C}_{T_{t,s}} = \frac{\partial \mathcal{C}_t}{\partial T_s}$:

$$d\boldsymbol{y} = \mathbf{C}_y d\boldsymbol{y} + \mathbf{C}_T d\boldsymbol{T} + d\boldsymbol{g}$$

- Recall that after-tax income of each household was proportional to $y_t - T_t$. We thus have $\mathbf{M} := \mathbf{C}_y = -\mathbf{C}_T$.

INTERTEMPORAL KEYNESIAN CROSS

- We can write the system of linearized equilibrium conditions as

$$d\boldsymbol{y} = \mathbf{M}\left(d\boldsymbol{y} - d\boldsymbol{T}\right) + d\boldsymbol{g},$$

- Auclert, Rognlie, and Straub (2024) call this system the Intertemporal Keynesian Cross (IKC).
- Solving for $d\boldsymbol{y}$:

$$d\boldsymbol{y} = \left(\mathbf{I} - \mathbf{M}\right)^{-1}\left(d\boldsymbol{g} - \mathbf{M}\, d\boldsymbol{T}\right).$$

⋆ *Note:* $\left(\mathbf{I} - \mathbf{M}\right)^{-1}$ is shorthand – $\mathbf{M}$ is an infinite-dimensional operator so standard matrix inversion does not apply; see ARS (2024) for a full argument.

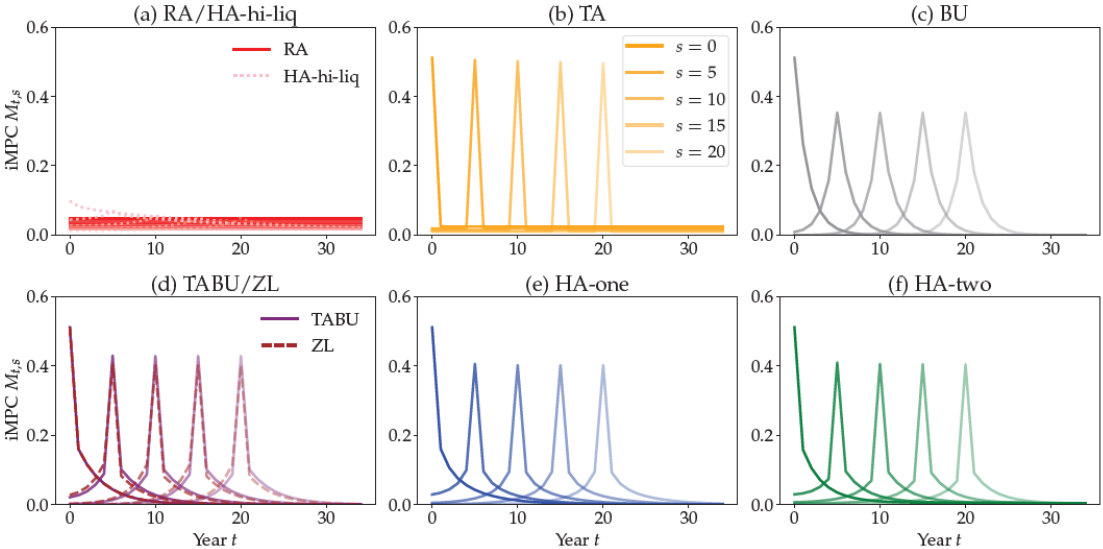
INTERTEMPORAL KEYNESIAN CROSS: **M** AS A SUFFICIENT STATISTIC

- We need to know "only" **M** to solve for the effect of fiscal policy $\boldsymbol{g}$, $\boldsymbol{T}$ on the economy.
- The matrix **M** is a sufficient statistic for the effect of fiscal policy.
- Example: two economies with different processes for idiosyncratic income, but with the same **M** $\rightarrow$ fiscal multipliers are the same.
- Challenge: **M** is an infinite-dimensional object – how to measure it?

INTERTEMPORAL MARGINAL PROPENSITY TO CONSUME

- **M** is a matrix of intertemporal marginal propensities to consume (iMPCs).
- It describes how aggregate consumption at t responds to changes in after-tax income at s.
- We established it is a sufficient statistic for the effect of fiscal policy in this class of models...
- ...but it is an infinite-dimensional object – how can we even measure it?

IMPCS ACROSS MODELS

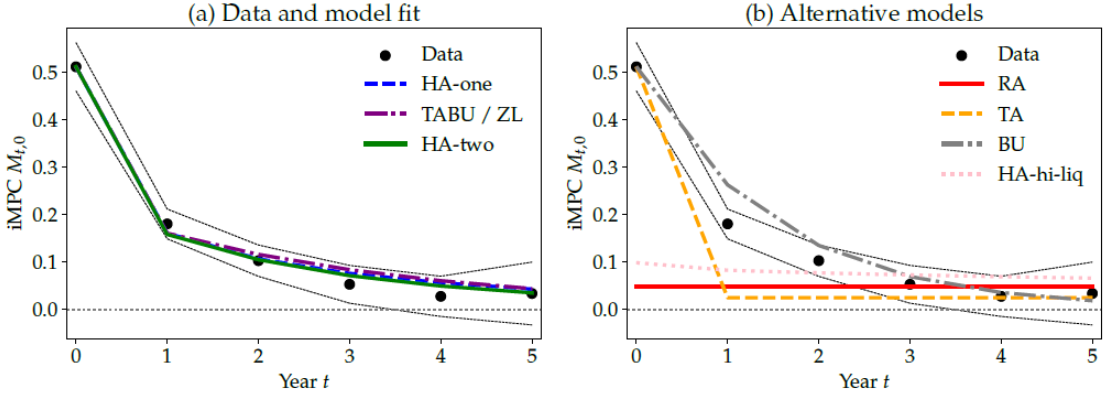


Source: Auclert et al. (2024)

ESTIMATING IMPCS FROM DATA

- **Norwegian data** (Fagereng, Holm, Natvik 2021): lottery winners identify the first column of **M**.
  - Since lottery wins are unforecastable, the regression of $c_{i,t+k}$ on lottery_{it} recovers M_{k0} (income-weighted average of individual responses).
  - Finding: $M_{00} \approx 0.51$, $M_{10} \approx 0.18$, slowly decaying thereafter.
- **Italian data** (SHIW 2016): households report their own MPC out of a windfall.
  - Income-weighted average of survey MPCs identifies M_{00} directly (≈ 0.44).
  - For $t > 0$: ARS construct a lower bound on M_{t0} using $(1+r)(1-\text{MPC}_i) \cdot \text{MPC}_i$ weights, under the assumption that the MPC distribution is stationary across years.
- Both series are closely aligned (see next slide).
- **Limitation**: these data identify only the first column of **M**. Other columns (anticipated income) are hard to measure. Fully constructing **M** requires additional model structure.

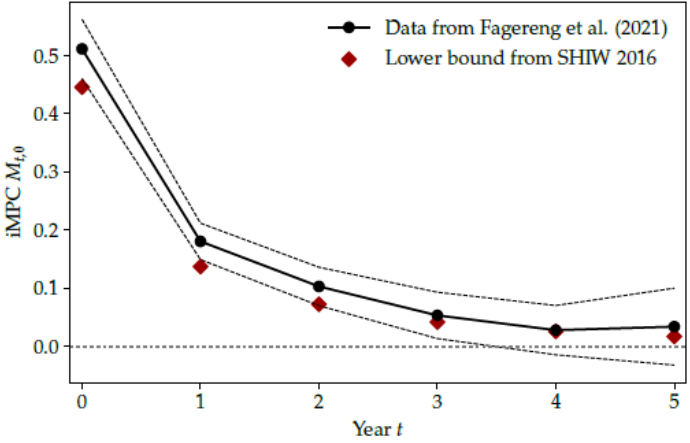
IMPCS IN THE DATA



Source: Auclert et al. (2024)

IMPCS: MODEL VS. DATA

Figure 1: iMPCs in the Norwegian and Italian data



Source: Auclert et al. (2024)

BALANCED BUDGET

- In an old-fashioned Keynesian model, for $dg = dT$ we have $dy = dg$.
- Same holds in the IKC with $d\boldsymbol{g} = d\boldsymbol{T}$:

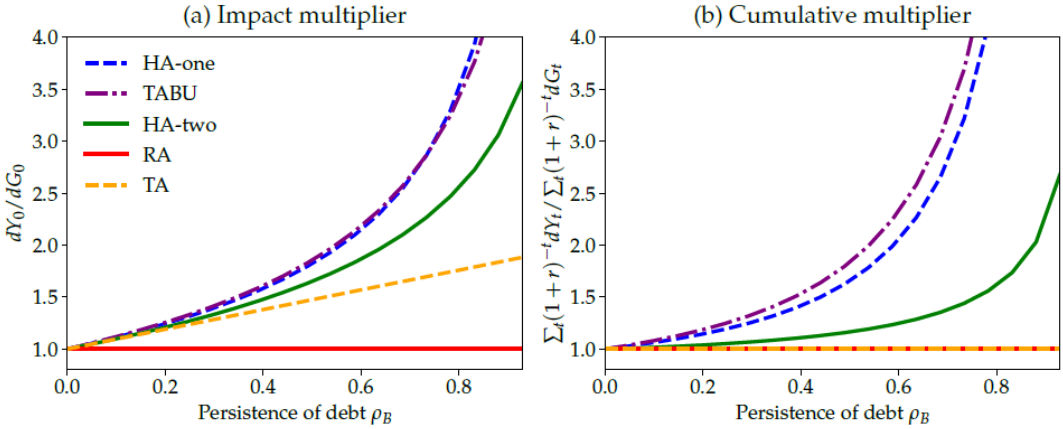
$$
\begin{aligned}
d\boldsymbol{y} &= \mathbf{M}d\boldsymbol{y} - \mathbf{M}d\boldsymbol{T} + d\boldsymbol{g} \\
&= \mathbf{M}d\boldsymbol{y} - \mathbf{M}d\boldsymbol{g} + d\boldsymbol{g} \\
&= \mathbf{M}d\boldsymbol{y} + (\mathbf{I} - \mathbf{M})\, d\boldsymbol{g} \\
\implies (\mathbf{I} - \mathbf{M})\, d\boldsymbol{y} &= (\mathbf{I} - \mathbf{M})\, d\boldsymbol{g} \\
\implies d\boldsymbol{y} &= d\boldsymbol{g}
\end{aligned}
$$

- This is a general property of the IKC. It holds irrespective of the details of the model, as long as we can write it in this form.

DEFICIT-FINANCED SPENDING

- **Balanced-budget** ($d\boldsymbol{g} = d\boldsymbol{T}$): cumulative multiplier = 1 always (as shown earlier).
- **Deficit-financed** spending: output response depends on iMPCs and the primary deficit path.
- RANK: Ricardian equivalence ⇒ deficits have no consumption effect ⇒ cumulative multiplier = 1.
- TANK: HtM agents boost the impact multiplier, but they spend only in the current period ⇒ cumulative multiplier = 1. As deficits turn into surpluses, consumption declines and the stimulus is fully reversed.
- HANK: households gradually spend down excess assets accumulated during deficits, creating persistent income-spending feedback via off-diagonal iMPCs $M_{t,s}$ ($t > s$). Cumulative multiplier > 1.

MULTIPLIERS



Source: Auclert et al. (2024)